THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VII — MATHEMATICS · ANNUAL / FULL-YEAR PRACTICE PAPER

Session 2026-27 | CBSE / RBSE Pattern

Class: 7	Subject: Mathematics	Set 03	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. This question paper contains 16 questions. All questions are compulsory.
2. Question paper is divided into FIVE sections – Section 'A', 'B', 'C', 'D' and 'E'.
3. In Section 'A' question number 1 has 12 multiple choice sub-parts carrying 1 mark each.
4. In Section 'B' question numbers 2 to 7 are very short answer type questions carrying 2 marks each.
5. In Section 'C' question numbers 8 to 10 are short answer (SA) type questions carrying 3 marks each.
6. In Section 'D' question numbers 11 to 13 are long answer (LA) type questions carrying 5 marks each.
7. In Section 'E' question numbers 14 to 16 are case-based questions whose sub-parts are of 1, 1 and 2 marks respectively.
8. There is no overall choice. However, internal choice has been provided in 1 question of Section B, 1 question of Section C, 2 questions of Section D, and in the 2-mark sub-part of every case-based question in Section E. Attempt only one of the alternatives in all such questions.
9. Draw neat figures wherever required. Use of calculators is NOT allowed. Write the serial number of the question before attempting it.

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) The smallest 8-digit number is: **1**

(a) 10,00,000 (b) 99,99,999 (c) 1,00,00,000 (d) 1,00,000

(ii) How many zeros are there in one crore? **1**

(a) 7 (b) 5 (c) 6 (d) 8

(iii) The value of 100 − 20 × 3 + 4 ÷ 2 is: **1**

(a) 242 (b) 82 (c) 44 (d) 42

(iv) 3 hundredths + 5 thousandths equals: **1**

(a) 0.35 (b) 0.0035 (c) 0.08 (d) 0.035

(v) The product 0.8 × 0.5 is: **1**

(a) 4 (b) 0.4 (c) 0.04 (d) 1.3

(vi) Which of the following is a like term of 5xy? **1**

(a) 5x (b) 5y (c) x^2y (d) 7xy

(vii) The perimeter of a square of side 2.5 cm is: **1**

(a) 6.25 cm (b) 10 cm (c) 5 cm (d) 12.5 cm

(viii) The supplement of an angle of 125° is: **1**

(a) 35° (b) 55° (c) 65° (d) 235°

(ix) Two lines intersect so that vertically opposite angles are (5x − 20)° and (3x + 10)°. The value of x is: **1**

(a) 5 (b) 10 (c) 15 (d) 30

(x) Which of these numbers is divisible by 9? **1**

(a) 738 (b) 734 (c) 736 (d) 740

(xi) A triangle can have at most how many obtuse angles? **1**

(a) Two (b) Three (c) None (d) One

(xii) How many pieces of 1/6 m can be cut from a ribbon 2 1/2 m long? **1**

(a) 15 (b) 12 (c) 5/12 (d) 25

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Round 7,86,42,350 (i) to the nearest crore and (ii) to the nearest lakh. **2**

3. Evaluate: 2 × [30 − (4 + 6) ÷ 2]. **2**

4. A bus travels at a steady speed of 42.5 km per hour. How far does it go in 3.2 hours? **2**

OR

Find 0.36 × 0.25.

5. An auto-rickshaw charges a fixed ₹ 30 plus ₹ 12 for each kilometre. Write an expression for the fare for k km and find the fare for 7 km. **2**

6. Four angles meet at a point O. Three of them measure 90°, 110° and 100°. Find the fourth angle. **2**

7. Write the first eight terms of the Virahanka (Fibonacci) type sequence that starts with 1, 2, and in which each term is the sum of the two terms before it. **2**

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. A tank of capacity 90 litres is 2/3 full. 3/4 of the water in it is used for watering plants. How much water is left in the tank? **3**

OR

Divide: 3 3/4 ÷ 5/8. Verify your answer by multiplication.

9. In triangle PQR, ∠Q = 2 × ∠P and ∠R = ∠P + 60°. Find the three angles. What kind of triangle is it? **3**

OR

Two sides of a triangle are 4 cm and 5 cm. Between which two numbers must the third side lie? What is the greatest whole-number perimeter possible?

10. Petrol costs ₹ 105.40 per litre. Rohit buys 5.5 litres. (i) Find the cost. (ii) He pays with a ₹ 600 note. How much change does he get? **3**

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

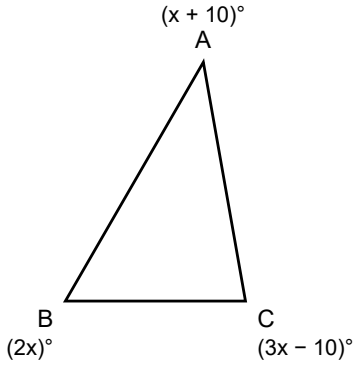
Questions 11 to 13 carry 5 marks each.

11. Simplify each. (i) 2/3 × 3/8 (ii) 4/5 ÷ 2/15 (iii) 1 1/2 × 2 2/3 (iv) Find the reciprocal of 7/9 and the product of the number and its reciprocal. (v) How many pieces of cloth, each 3/4 m long, can be cut from a roll of 9 m? **5**

OR

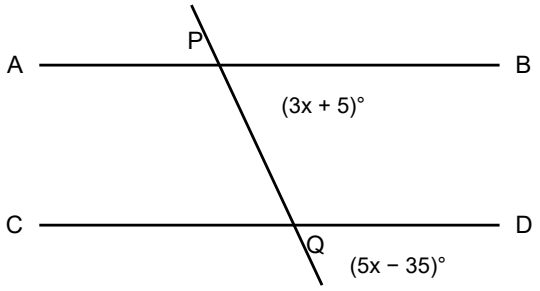
Rahul reads 2/5 of a book on Monday and 1/3 of the remaining part on Tuesday. (i) What fraction of the book remains after Monday? (ii) What fraction of the whole book does he read on Tuesday? (iii) What fraction is read in all? (iv) What fraction remains unread? (v) If 120 pages remain unread, how many pages does the book have?

12. In the figure, the angles of triangle ABC are (x + 10)°, (2x)° and (3x − 10)°. (i) State the angle sum property. (ii) Find x. (iii) Find the three angles. (iv) Is the triangle acute, right or obtuse? (v) Find the exterior angle formed when side BC is extended at C (assume ∠C = (3x − 10)°). **5**



OR

In the figure, AB ∥ CD and PQ is a transversal. ∠BPQ = (3x + 5)° and the corresponding angle at Q below CD is (5x − 35)°. (i) Write the relation between the two angles. (ii) Find x. (iii) Find the measure of the angle. (iv) Find ∠DQP. (v) Find ∠CQP.



13. (i) Write the sum of four consecutive even numbers n, n + 2, n + 4, n + 6 in its simplest form. (ii) Show that it is always divisible by 4. (iii) Check it for n = 10. (iv) A two-digit number has tens digit a and units digit b. Write it, and write the number with digits reversed. (v) Show that the sum of a two-digit number and its reverse is a multiple of 11; verify with 47. **5**

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. The Narendra Modi Stadium in Ahmedabad is one of the largest cricket stadiums in the world and can seat about 1,32,000 spectators. Suppose each ticket for a match costs ₹ 800.

(i) Write 1,32,000 in the international system. 1

(ii) How many lakh is 1,32,000? 1

(iii) Find the money collected if all the seats are sold at ₹ 800 each. Write the answer in the Indian system and in crore. 2

OR

(iii) About how many such stadiums would be needed to seat 1 crore people?

15. Rani has 5/6 kg of dal. She uses 3/5 of it to cook lunch.

(i) How much dal does she use for lunch? 1

(ii) How much dal is left? 1

(iii) She packs the remaining dal into packets of 1/12 kg each. How many packets does she fill? 2

OR

(iii) Her neighbour gives her 2/3 kg more dal. How much dal does she have now?

16. Tara notices a pattern: 11 × 11 = 121, 111 × 111 = 12321, 1111 × 1111 = 1234321. All the answers read the same forwards and backwards.

(i) What is the name for such numbers? 1

(ii) Predict 11111 × 11111. 1

(iii) Is 12321 divisible by 9? Use the divisibility rule and verify by division. 2

OR

(iii) Is 1234321 divisible by 3? Give a reason.